SD Model

Data preparation

The training data contain 89 observations, three Re levels and three Fr levels. The combination Re = 398, Fr = 0.3 is absent.

EDA: motivate transformations and flexible relationships

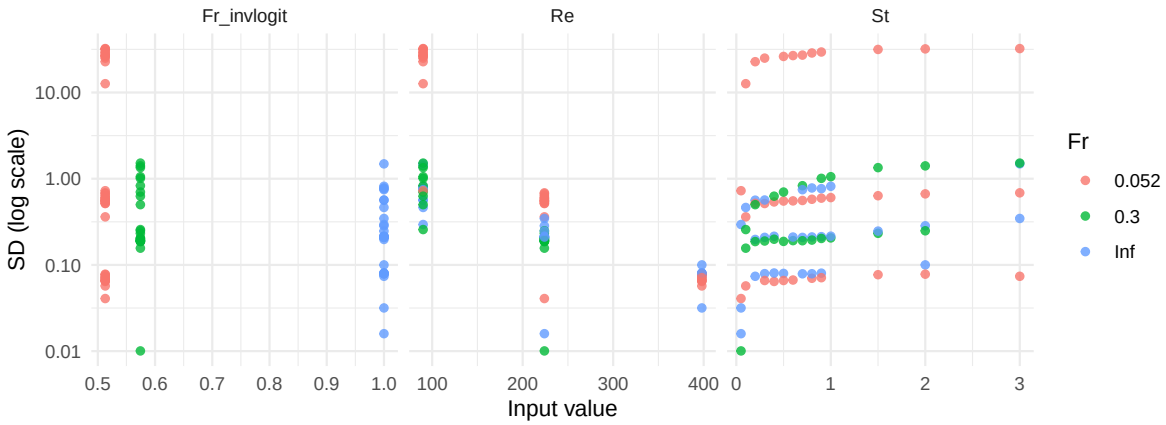


Figure 1: SD across the inputs. Only three Re and Fr levels are observed; curvature between these levels is model-imposed.

Candidate models and validation

Compare predictive performance

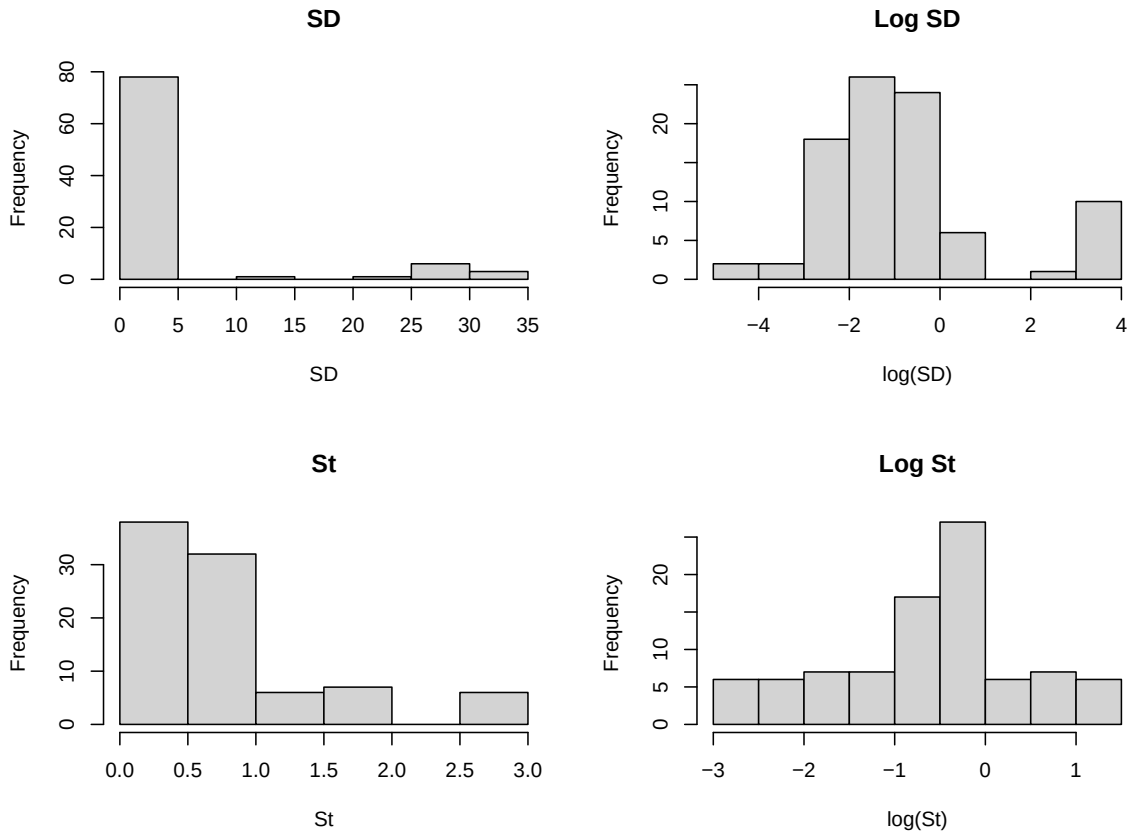


Figure 2: Original and log scales for SD and St, used to assess transformation candidates.

Table 1: LOOCV prediction errors on the original SD scale (lower is better).

Model	RMSE	MAE
Second-order Box-Cox model	2.51	0.82
Categorical log model: Re-Fr interaction	2.90	0.98
Second-order Box-Cox + LASSO	3.17	1.08
Second-order log model	3.84	1.59
Second-order log + LASSO	3.99	1.65
Continuous log model: Re-Fr interaction	7.34	3.27
Continuous Box-Cox: Re-Fr interaction	7.95	3.34

The lowest observed LOOCV RMSE is from the **Second-order Box-Cox model** (RMSE 2.51, MAE 0.82). Box–Cox alone does not improve the simple continuous model. Adding curvature and interactions together with Box–Cox produces the strongest candidate; LASSO does not improve on the corresponding unpenalized second-order models. All continuous candidates produced finite, positive LOOCV predictions.

Selected model and interpretation

For $x = \log(St)$, $r = Re/100$ and $g = \text{plogis}(Fr)$, the selected model is

$$\frac{\text{SD}^{\lambda} - 1}{\lambda} = \beta_0 + \beta_1 x + \beta_2 r + \beta_3 g + \beta_4 x^2 + \beta_5 r^2 + \beta_6 g^2 + \beta_7 xr + \beta_8 xg + \beta_9 rg + \varepsilon.$$

This is a second-order polynomial regression with a Box–Cox-transformed response. The full-data estimate is $\widehat{\lambda} =$ -0.110.

Table 2: Selected model coefficients, fitted to all training observations.

Term	Estimate
Intercept	42.525
x	0.240
r	-3.513
g	-107.581
x^2	-0.240
r^2	0.161
g^2	66.515
x*r	-0.041
x*g	-0.122
r*g	1.831

With squares and interactions, an input's effect depends on its value and the other inputs. Coefficients are on the Box–Cox scale; they are not direct percentage changes in SD. The curves below show the combined fitted effects on the original scale, using the same residual-based back-transformation as the predictions. They describe model-based associations. The fitted curves rise over lower St values and flatten or turn downward toward the upper observed range. Differences between Fr conditions also vary across Re.

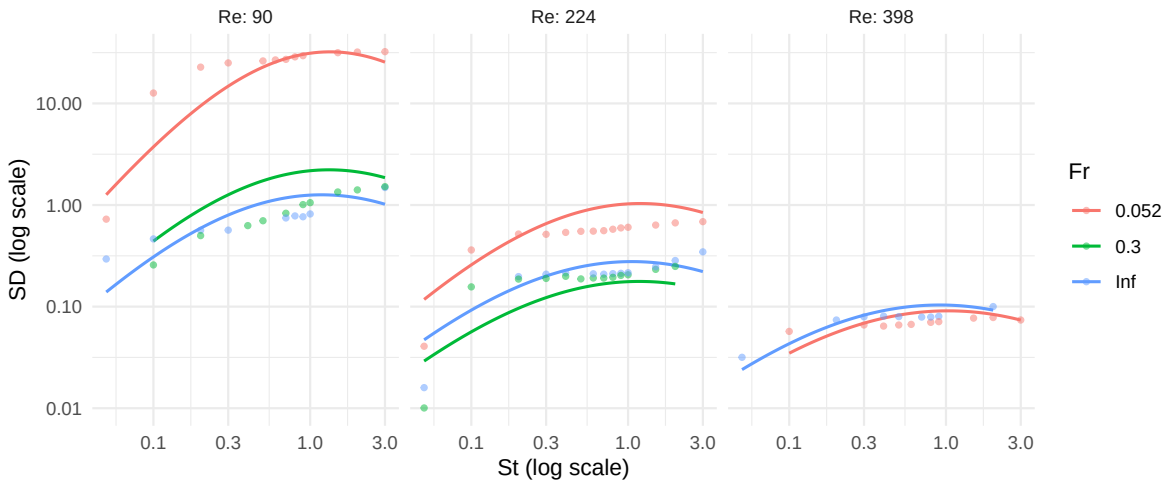


Figure 3: Full-data fitted curves and observed SD for sampled Re-Fr combinations. These are descriptive fits, not held-out predictions or uncertainty intervals.

Prediction performance and model diagnostics

The prediction plot assesses errors on observations excluded from fitting. The following diagnostics instead examine the full-data model on the Box–Cox response scale.

The second-order Box–Cox model is the leading SD surrogate candidate in this comparison. Improvements support flexible predictor relationships and response transformation more than shrinkage. The residual trend retains curvature, and the Q-Q plot shows tail departures, so lower prediction error does not establish that all regression assumptions hold. Influential simulations are retained as valid observations.

Appendix: earlier model searches

These tables document the initial staged searches. They explain the benchmark and simple continuous specification without repeating their diagnostics.

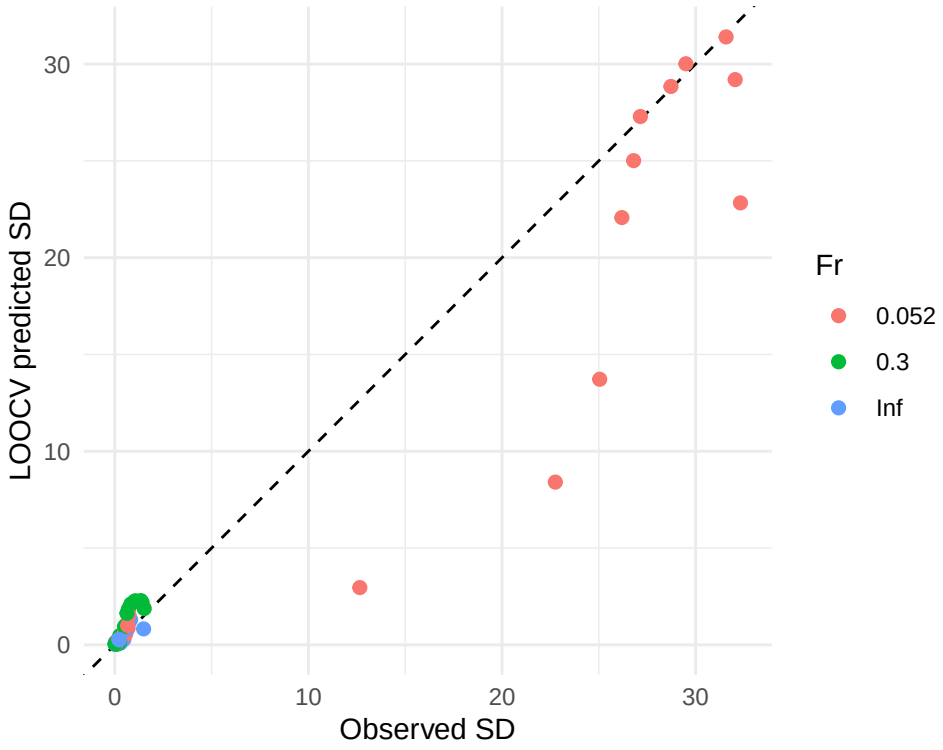


Figure 4: Held-out predictions for the selected candidate. The dashed line represents perfect prediction.

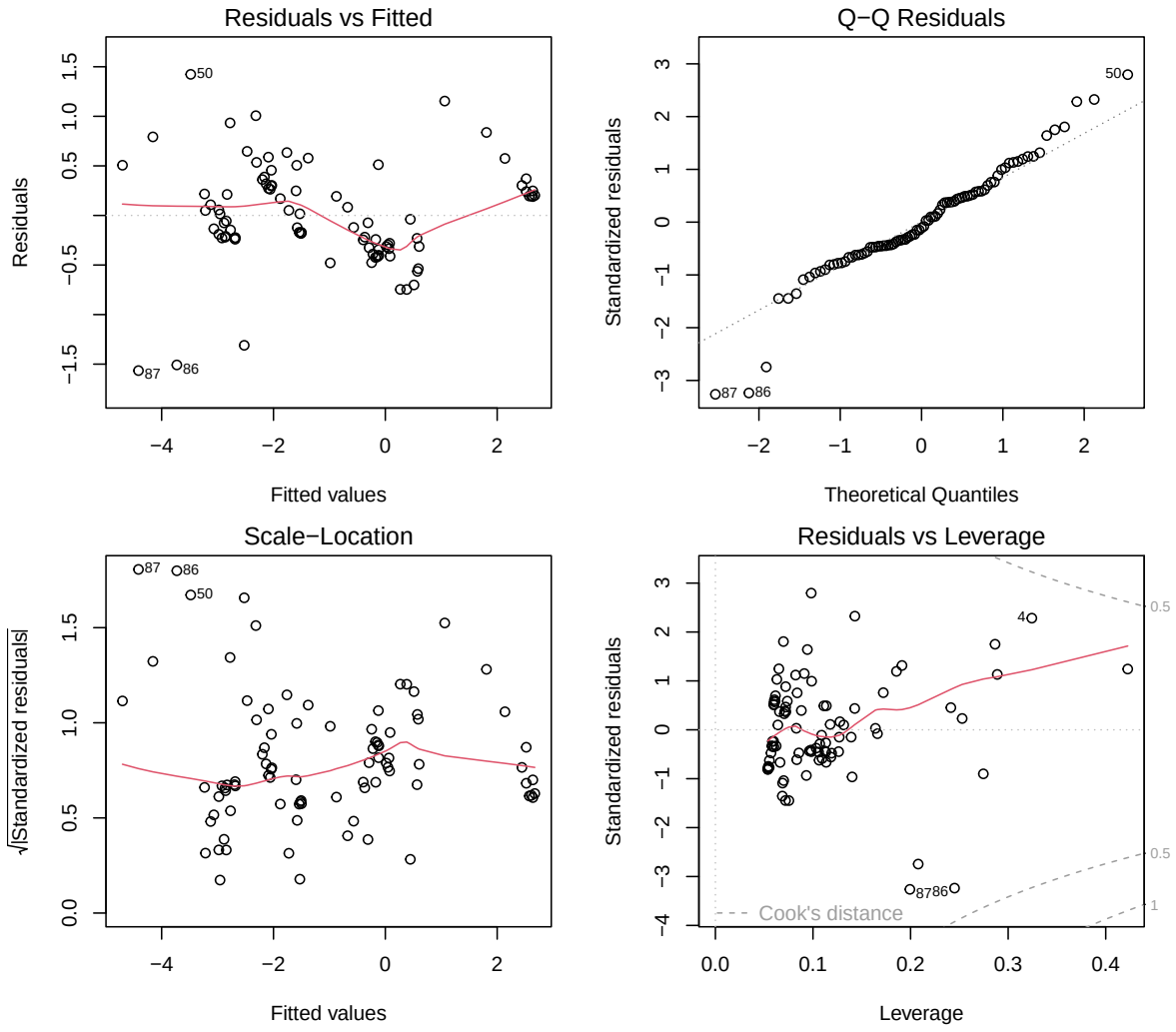


Figure 5: Selected-model diagnostics: residual structure, normal Q-Q, changing spread, and influence. Box-Cox estimation is treated as fixed for these descriptive checks.

Table 3: Categorical inputs: additive log-transformation search.

model	RMSE	MAE
log_both	5.99	2.52
log_SD	6.36	2.65
log_St	6.68	5.72
no_logs	6.77	5.79

Table 4: Categorical inputs: interaction search after selecting transformations.

model	RMSE	MAE
Re_Fr	2.90	0.98
pairwise	3.99	1.25
St_Re	5.81	2.48
additive	5.99	2.52
St_Fr	6.10	2.59
threeway	6.91	1.77

Table 5: Continuous inputs: additive log-transformation search.

model	RMSE	MAE
log_both	7.82	3.63
log_St	7.93	5.83
log_SD	7.97	3.68
no_logs	8.02	5.85

Table 6: Continuous inputs: interaction search after selecting transformations.

model	RMSE	MAE
Re_Fr	7.34	3.27
pairwise	7.35	3.30
threeway	7.36	3.31
St_Re	7.79	3.66
St_Fr	7.82	3.64
additive	7.82	3.63

For Box–Cox plus LASSO, the transformation parameter is estimated from the corresponding unpenalized design, rather than jointly with the penalty. The full-data penalty is at the

weakest end of the tested grid and retains all predictor terms. This limits claims about an optimal penalty or variable selection.

Table 7: Penalty-grid boundary check for Box-Cox plus LASSO.

outer_folds	weakest_penalty_folds
89	65